

R code for Data Science for Beginners

Day 1: Individual Exercise

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Task 0. Getting things ready for programming

If you want to render a pdf as your quarto output, you would first need to download Latex first:

Install Latex on your computer:

- Mac: [here](#)
- Windows: [here](#)

Please read the following links carefully:

- [Getting started with Quarto](#)
- [Markdown Basics](#)
- [File naming system](#)

Task 1. Modify the author name above to your name

- Completed. Please view above.

Task 2. Perform the following calculations by writing R commands

```
### 2-1-Example: Multiply 4 by 30 plus 6, and then raise the answer to the  
###power of 0.5  
(4 * (30 +6)) ^ 0.5 # a sample answer is here
```

```
[1] 12
```

```
### 2-1-Attempt: Multiply 3 by 20 minus 6, and then raise the answer to the
###power of 2
(3 *(20-6))^2 #I was not sure if I needed to create an additional equation.
```

```
[1] 1764
```

```
###I did just in case.
```

```
### 2-2: Multiply 4 by 30, then add 6 and then raise the answer to the power
###of 0.5
```

```
((4*30)+6)^0.5
```

```
[1] 11.22497
```

Note that 2-1 and 2-2 should give you DIFFERENT answers, because these are different operations.

Task 3. Working with objects

3-1: Create an object called "X1" which is the number 73

```
### 3-1: Create an object called "X1" which is the number 73
x1=73
x1
```

```
[1] 73
```

3-2: Create another object called "X2" which is the answer to the sum $99 + 38$

```
### 3-2: Create another object called "X2" which is the answer to the sum 99 +
###38
x2=99+38
x2
```

```
[1] 137
```

3-3: Multiply X1 and X2 together and store the answer as another object called "X3"

```
### 3-3: Multiply X1 and X2 together and store the answer as another object  
##called "X3"  
x3=x1*x2  
x3
```

```
[1] 10001
```

3-4: Subtract 1 from X3, and then raise the answer to the power of 0.25

```
### 3-4: Subtract 1 from X3, and then raise the answer to the power of 0.25  
(x3-1)^0.25
```

```
[1] 10
```

Note: the answer should be 10.

```
###Note: the answer should be 10.  
if((x3-1)^0.25==10){  
  print("The Answer is 10.")  
}
```

```
[1] "The Answer is 10."
```

Task 4. Calculation using objects

Redo the calculation in 2-1 (Multiply 4 by 30 plus 6, and then raise the answer to the power of 0.5), but this time do so by creating objects. Take the following steps:

4-1: Create an object called part.1 that is equal to "30 plus 6"

```
### 4-1: Create an object called part.1 that is equal to "30 plus 6"  
part.1=30+6  
part.1
```

```
[1] 36
```

4-2: Create an object called `part.2` that is equal to 4 times `part.1`

```
### 4-2: Create an object called part.2 that is equal to 4 times part.1
part.2=part.1*4
part.2
```

```
[1] 144
```

4-3: Raise `part.2` to the power of 0.5

```
### 4-3: Raise part.2 to the power of 0.5
answerQuestion4=part.2^0.5
answerQuestion4
```

```
[1] 12
```

The numerical answer you get here should be the same as your answer for 2.1

```
if(answerQuestion4==(4 * (30 +6)) ^ 0.5) {
  print("The two answers are the same. 12.")
}
```

```
[1] "The two answers are the same. 12."
```

Task 5. Detecting R data types

Now, you have seen from the above examples how you should generated your answers and results in `.qmd` and `.rmd` documents. Please continue and finish the following questions.

5-1 Create a character vector and show that it is character. It'd be nice if you can do it in one line of code

```
### 5-1 Create a character vector and show that it is character. It'd be nice
###if you can do it in one line of code
class(c("Hello", "I", "Like", "To", "Play", "Minecraft"))
```

```
[1] "character"
```

5-2 Create a numeric vector and coerce it to a factor vector. It'd be nice if you can do it in one line of code

```
### 5-2 Create a numeric vector and coerce it to a factor vector. It'd be nice  
###if you can do it in one line of code
```

```
class(as.factor(c(1,2,3,4,5,6,7,8,9)))
```

```
[1] "factor"
```

Task 6. Testing NA

6-1 Create a numeric vector with NA

```
### 6-1 Create a numeric vector with NA  
partSixVector=c(2,4,6,8,9, NA, 15,16,1553)  
partSixVector
```

```
[1] 2 4 6 8 9 NA 15 16 1553
```

6-2 Show me where the NA is located in the vector (using the which() function)

```
### 6-2 Show me where the NA is located in the vector (using the which()  
###function)  
which(is.na(partSixVector))
```

```
[1] 6
```

6-3 Create a numeric vector with NaN

```
### 6-3 Create a numeric vector with NaN  
partSixVector2=c(2,4,6,8,9, (0/0), 15,16,1553)  
partSixVector2
```

```
[1] 2 4 6 8 9 NaN 15 16 1553
```

6-4 Show me where the NaN is located in the vector (using the which() function)

```
### 6-4 Show me where the NaN is located in the vector (using the which()  
###function)  
which(is.nan(partSixVector2))
```

```
[1] 6
```

Finally, execute the entire file. Make sure that you don't get any error messages, such as "Error: unexpected symbol in ..." shown in Red. If you receive an error message, it's probably because you forgot to comment out some lines. Comment them out so you won't get any error message.